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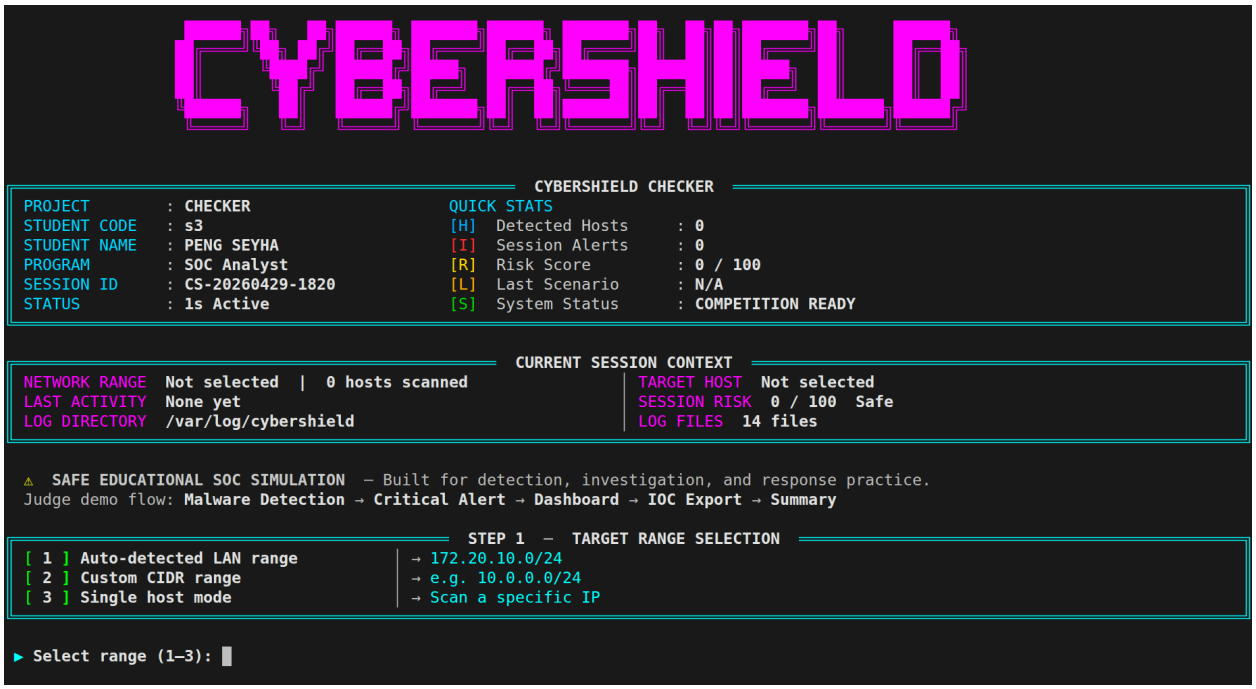
1. Introduction



CyberShield Simulator is a Bash-based project developed in Kali Linux to simulate Security Operations Center (SOC) activities in a safe and controlled environment. The purpose of this project is to demonstrate how SOC analysts detect, analyze, and respond to cybersecurity threats. The system performs network scanning, identifies active hosts, and simulates various attack scenarios. It generates SOC alerts that include incident IDs, severity levels, MITRE ATT&CK mapping, and risk scores. All activities are logged for further analysis and investigation.

This project is designed strictly for educational purposes. It does not perform real harmful attacks, but instead replicates realistic attack behaviors to improve understanding of SOC workflows, incident response, and security monitoring practices.

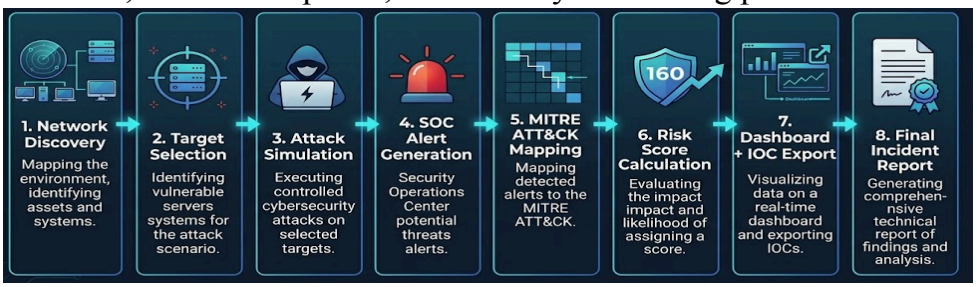


***Figure 1**. CyberShield SOC Simulator process workflow.*

2. Project Overview

CyberShield Simulator provides a practical environment for learning SOC operations. It allows users to scan networks, select targets, simulate attacks, and analyze generated alerts. The system produces realistic SOC outputs, including:

- Incident identification
- Severity classification
- Risk scoring
- MITRE ATT&CK mapping
- Analyst investigation steps

3. Tools and Technologies

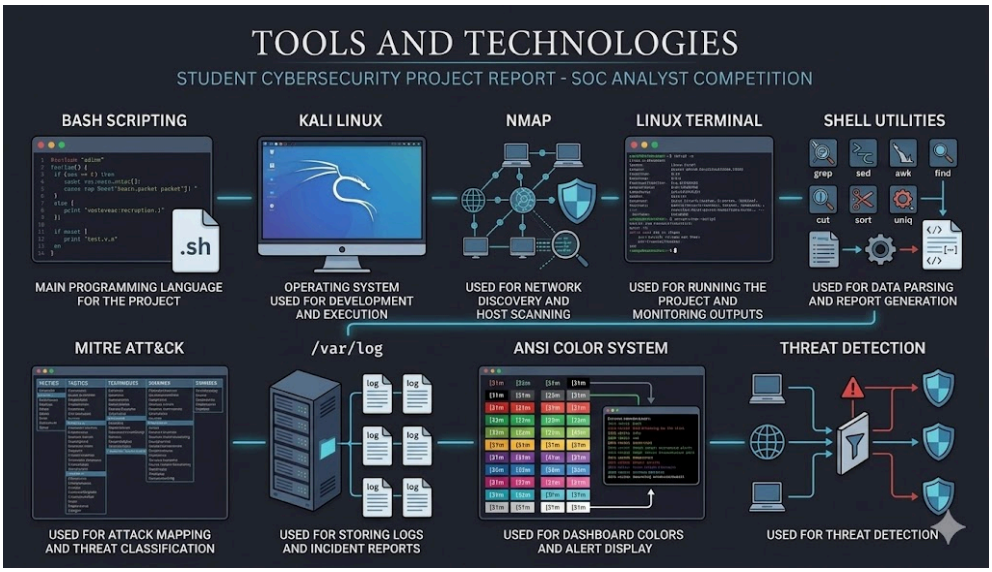


Figure 1. *Tools and technologies used in the CyberShield Simulator project.*

- Bash Scripting – main programming language for the project
- Kali Linux – operating system used for development and execution
- Nmap – used for network discovery and host scanning
- Linux Terminal – used for running the project and monitoring outputs
- /var/log – used for storing logs and incident reports
- MITRE ATT&CK – used for attack mapping and threat classification
- ANSI Color System – used for dashboard colors and alert display
- Shell Utilities – used for data parsing and report generation

4. System Architecture and Implementation

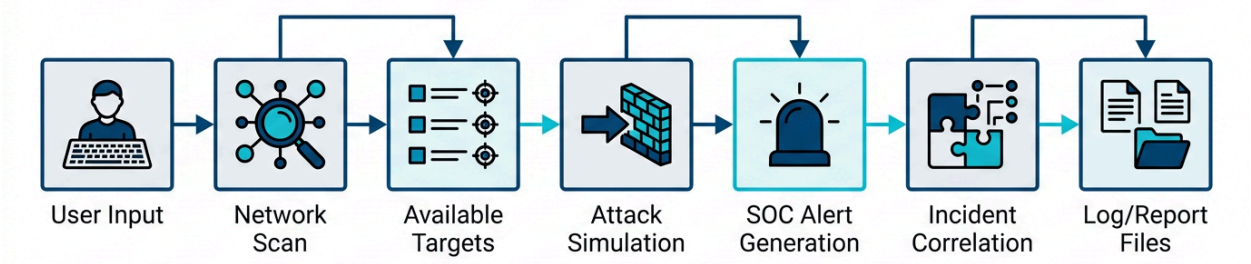


Figure 2. *System architecture and workflow of the CyberShield SOC Simulator.*

4.1 Environment Setup

Before starting the simulation, the system prepares the working environment required for the project. Since CyberShield Simulator is developed using Bash Script in Kali Linux, the terminal environment must be ready first.

The system automatically creates the required folders and log files inside **/var/log/cybershield/** to store incidents, reports, IOC exports, and session summaries. This is important because every simulation must be recorded for later investigation and final reporting.

```
┌──(kali㉿kali)-[~]
└─$ sudo tree /var/log/cybershield/
/var/log/cybershield/
├── correlation.log
├── incidents.log
├── ioc_export.txt
├── reports.log
└── summary.log
```

Figure 3*. Log directory structure of CyberShield Simulator showing incident logs, reports, and IOC export files.*

```
setup_log_paths() {
    if [ -w /var/log ] 2>/dev/null || [ "$(id -u)" -eq 0 ]; then
        LOG_DIR="/var/log/cybershield"
    else
        LOG_DIR="$HOME/.cybershield/logs"
    fi
    mkdir -p "$LOG_DIR" 2>/dev/null || { echo "Cannot create log directory: $LOG_DIR"; exit 1; }
    chmod 700 "$LOG_DIR" 2>/dev/null || true
    LOG_FILE="$LOG_DIR/incidents.log"
    REPORT_FILE="$LOG_DIR/reports.log"
    SUMMARY_FILE="$LOG_DIR/summary.log"
    CORRELATION_FILE="$LOG_DIR/correlation.log"
    IOC_FILE="$LOG_DIR/ioc_export.txt"
    local f
    for f in "$LOG_FILE" "$REPORT_FILE" "$SUMMARY_FILE" "$CORRELATION_FILE" "$IOC_FILE"; do
        touch "$f" 2>/dev/null || { echo "Cannot create: $f"; exit 1; }
        chmod 600 "$f" 2>/dev/null || true
    done
}
```

Code Snippet1. *Environment setup function for preparing CyberShield Simulator log files.*

4.2 Network Range Selection

After the environment setup, the system allows the user to select the target network range for scanning. Three options are provided: auto-detected LAN range, custom CIDR range, and single host mode.

This step helps define the scanning scope before host discovery begins. It makes the simulator more flexible and closer to real SOC investigation workflows.

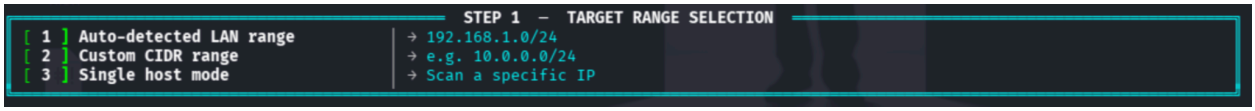


Figure 4. *Network range selection menu of CyberShield Simulator.*

```
select_network_range() {
    local range choice custom single
    range=$(get_current_network_range)
    CURRENT_NETWORK_RANGE="$range"
    show_network_menu "$range"
    read_menu_choice "Select range (1-3):" choice
    case "$choice" in
        1) if [ -n "$range" ]; then
               SELECTED_SCAN_RANGE="$range"
               status_ok "Range: ${C_WHITE}${range}${NC}"
           else
               status_error "Cannot auto-detect range."
               return 1
           fi ;;
        2) safe_read "Enter CIDR (e.g. 192.168.1.0/24):" custom
           if validate_range "$custom"; then
               SELECTED_SCAN_RANGE="$custom"
               status_ok "Custom range: ${C_WHITE}${custom}${NC}"
           else
               status_error "Invalid CIDR."
               return 1
           fi ;;
        3) safe_read "Enter host IP:" single
           if validate_ip "$single"; then
               SELECTED_SCAN_RANGE="${single}/32"
               status_ok "Single host: ${C_WHITE}${single}${NC}"
           else
               status_error "Invalid IP."
               return 1
           fi ;;
        *) status_error "Invalid choice."; return 1 ;;
    esac
}
```

Code Snippet 2. *Network range selection function used for selecting scan scope.*

4.3 Host Discovery and Target Selection

After selecting the network range, the system uses Nmap to discover active hosts on the network. This helps identify available devices that can be used as targets for simulation and investigation. The user can then select a target manually, use the current machine, enter a custom IP address, or allow random target selection. This makes the simulator more practical and similar to real SOC operations.

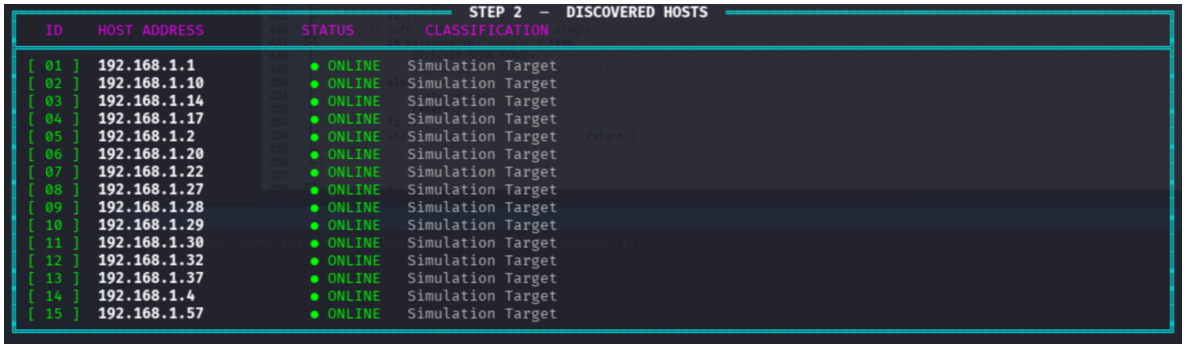


Figure 5. *Shows the host discovery result with active IP addresses detected by the system.*

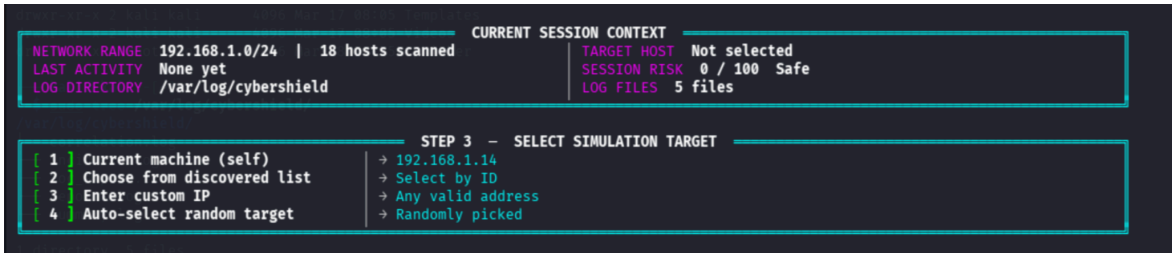


Figure 6*. Shows the target selection menu used for choosing the simulation target.*

4.4 Safe Attack Simulation Engine

After selecting a target, the system executes safe attack simulations such as Port Scanning, DoS behavior, Brute Force attempts, and Malware activity.

These simulations do not perform real attacks. Instead, they generate realistic behavioral patterns that trigger SOC alerts. Each alert includes:

- Unique incident ID
- Severity level
- MITRE ATT&CK mapping
- Risk score
- Recommended analyst response

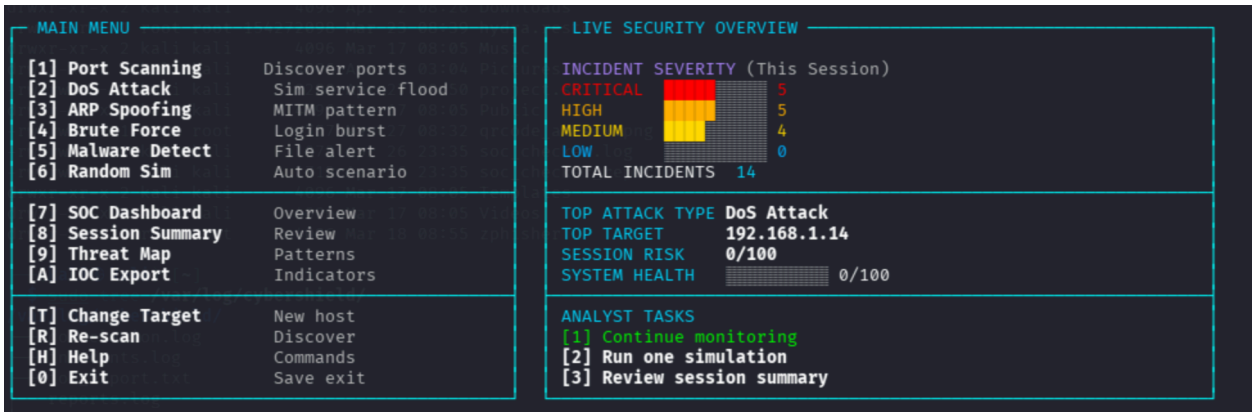


Figure 7. *Shows the attack selection menu used for choosing the simulation type.*

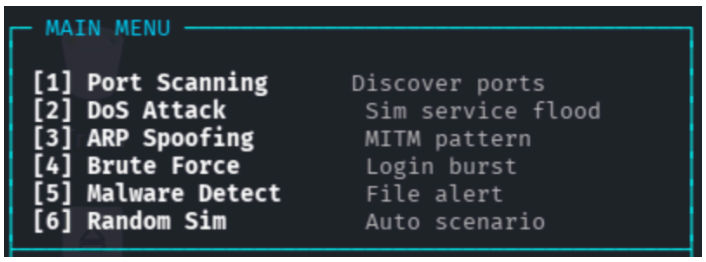
```
execute_attack() {
    local choice="$1" target="$2" actual_choice="$1"
    case "$choice" in
        1) sim_port_scan    "$target" ;;
        2) sim_dos          "$target" ;;
        3) sim_arp_spoof    "$target" ;;
        4) sim_brute_force  "$target" ;;
        5) sim_malware      "$target" ;;
        6)
            local r
            r=$((1 + RANDOM % 5))
            actual_choice="$r"
            status_info "Auto-selected module: ${C_WHITE}$(get_attack_name "$r")${NC}"
            case "$r" in
                1) sim_port_scan    "$target" ;;
                2) sim_dos          "$target" ;;
                3) sim_arp_spoof    "$target" ;;
                4) sim_brute_force  "$target" ;;
                5) sim_malware      "$target" ;;
            esac ;;
        *) status_error "Invalid module."; return 1 ;;
    esac
    SESSION_LAST_ATTACK=$(get_attack_name "$actual_choice")
}
```

Code Snippet 3. *Attack execution function used for running safe attack simulations .*

5. Attack Simulation Modules

The system simulates attacks such as Port Scanning, DoS attacks, Brute Force Login attempts, and Malware Detection. Instead of performing real harmful attacks, the simulator safely generates alerts, severity levels, MITRE ATT&CK mapping, risk scores, and analyst response steps.

Figure 8*. Attack simulation menu of CyberShield Simulator .*



5.1. Port Scanning Detection

Port scanning is commonly used by attackers to identify open services and potential vulnerabilities.

In this module, CyberShield simulates port scanning behavior and generates a SOC alert. The alert includes incident details, severity level, and risk score. The simulation was validated using Wireshark by observing TCP SYN packet activity in a controlled lab environment.

Figure 9. *Shows the Port Scanning alert generated by CyberShield Simulator.*

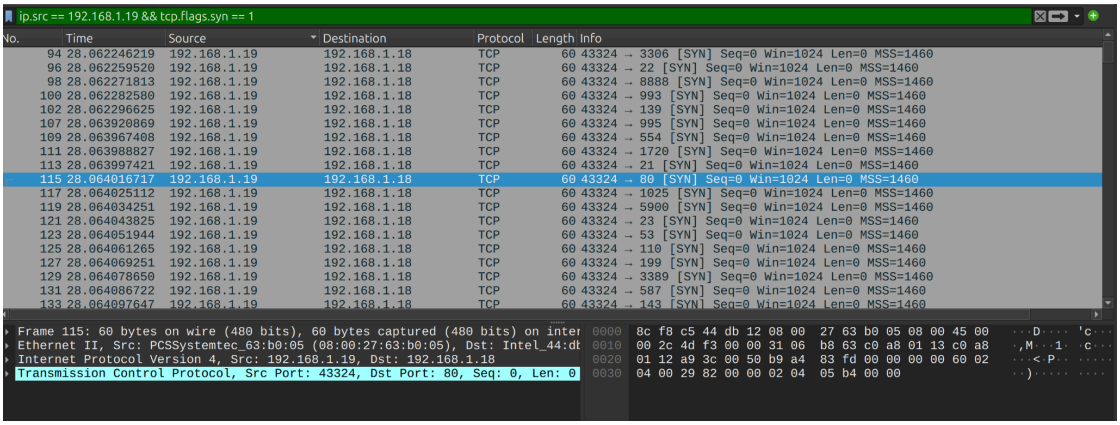


Figure 10. *Shows Wireshark packet capture during Port Scanning validation between Kali Linux and the Ubuntu target system.*

```
sim_port_scan() {
    local target="$1" ports sev
    ports=$((20 + RANDOM % 80))
    sev="Medium"; [ "$ports" -gt 60 ] && sev="High"
    status_warn "Port Scan on ${C_WHITE}${target}${NC}"
    ui_loading_bar "Enumerating services on ${target}" 22
    ui_box_top "$UI_WIDTH" "SIMULATION RESULTS — PORT SCAN"
    ui_metric_row "$UI_WIDTH" "Ports Probed"    "${C_WHITE}${ports}${NC}"  "Threshold"  "${C_WHITE}15${NC}"
    ui_box_line "$UI_WIDTH" "  ${C_SUCCESS}Exposed Services${NC}  ${C_PRIMARY}22/tcp ssh  80/tcp http  443/tcp https${NC}"
    ui_box_line "$UI_WIDTH" "  ${C_GRAY}Evidence${NC}          ${C_WHITE}${ports} probes exceeded detection threshold.${NC}"
    ui_box_bottom "$UI_WIDTH"
    create_soc_alert "Port Scanning" "$target" "$sev" \
        "Network IDS / Firewall Logs" "Open" "T1046 — Network Service Discovery" \
        "${ports} port probes exceeded threshold of 15." \
        "Pre-exploitation reconnaissance activity suspected." \
        "Review firewall rules and confirm scan authorization." \
        "Verify if source IP matches any authorized security scanner." \
        "Review firewall and IDS logs for source IP." \
        "Identify and close unnecessary listening ports." \
        "Escalate if source is unknown or scan repeats."
}
```

Code Snippet 5. *Shows the function used for simulating Port Scanning detection and generating the SOC alert.*

5.2. Dos Attack Simulation

Denial-of-Service (DoS) attacks attempt to overwhelm a system with excessive traffic, making it unavailable to legitimate users.

This module simulates abnormal traffic patterns and generates corresponding SOC alerts. The behavior was validated using repeated ICMP traffic observed through Wireshark in a controlled environment.

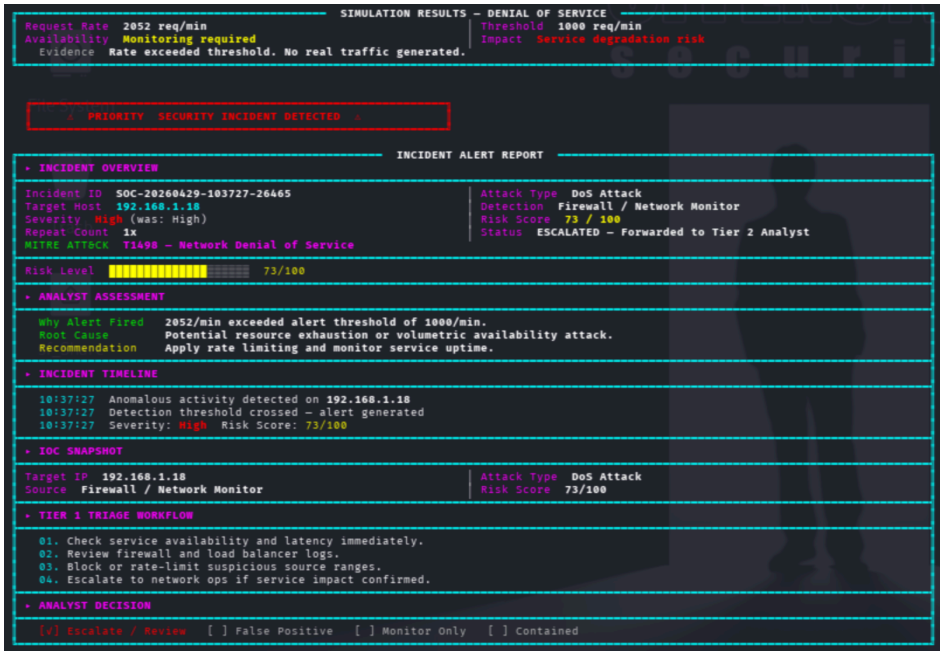


Figure 11. *Shows the DoS Attack alert generated by CyberShield Simulator.*

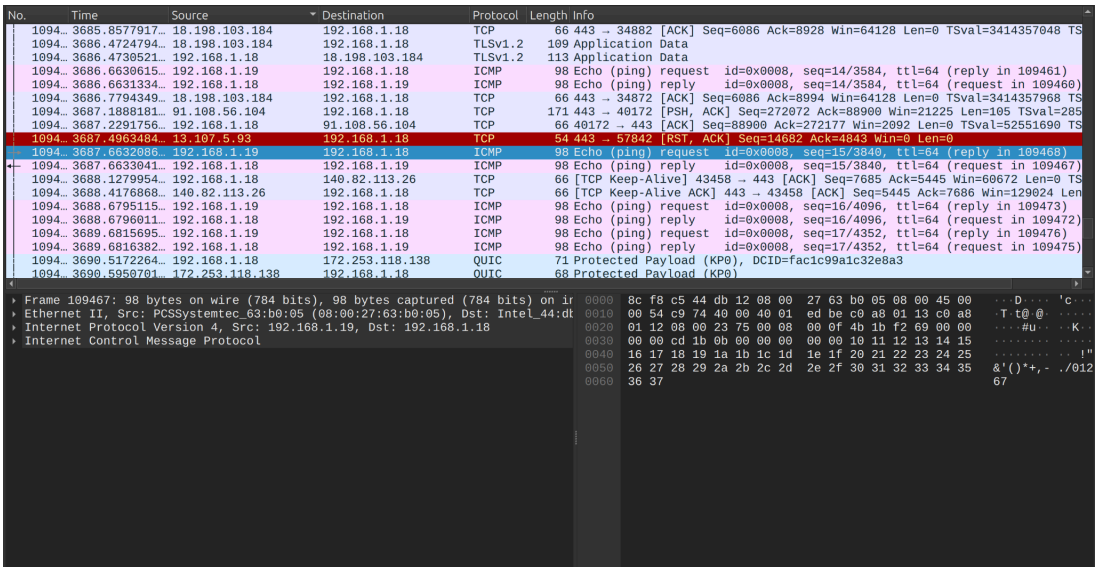


Figure 12. *Wireshark packet capture showing repeated ICMP traffic between Kali Linux and the Ubuntu target system during controlled DoS Attack simulation.*

```
}sim_dos() {
    local target="$1" rate sev
    rate=$((800 + RANDOM % 2500))
    sev="High"; [ "$rate" -gt 2500 ] && sev="Critical"
    status_warn "DoS Simulation on ${C_WHITE}${target}${NC}"
    ui_loading_bar "Analyzing traffic flood pattern on ${target}" 22
    ui_box_top "$UI_WIDTH" "SIMULATION RESULTS — DENIAL OF SERVICE"
    ui_metric_row "$UI_WIDTH" "Request Rate"  "${C_WHITE}${rate} req/min${NC}"      "Threshold"  "${C_WHITE}1000 req/min${NC}"
    ui_metric_row "$UI_WIDTH" "Availability"  "${C_WARN}Monitoring required${NC}"   "Impact"     "${C_DANGER}Service degradation ri
    ui_box_line "$UI_WIDTH" "  ${C_GRAY}Evidence${NC}  ${C_WHITE}Rate exceeded threshold. No real traffic generated.${NC}"
    ui_box_bottom "$UI_WIDTH"
    create_soc_alert "DoS Attack" "$target" "$sev" \
        "Firewall / Network Monitor" "Escalated — Tier 2" "T1498 — Network Denial of Service" \
        "${rate}/min exceeded alert threshold of 1000/min." \
        "Potential resource exhaustion or volumetric availability attack." \
        "Apply rate limiting and monitor service uptime." \
        "Check service availability and latency immediately." \
        "Review firewall and load balancer logs." \
        "Block or rate-limit suspicious source ranges." \
        "Escalate to network ops if service impact confirmed."
}
```

Code Snippet 5. *Shows the function used for simulating DoS detection and generating the SOC alert.*

5.3. Brute Force Login Detection

Brute force attacks involve repeated login attempts to gain unauthorized access.This module simulates multiple failed login attempts and triggers a SOC alert when suspicious activity is detected. The system generates incident details, severity classification, and investigation guidance.

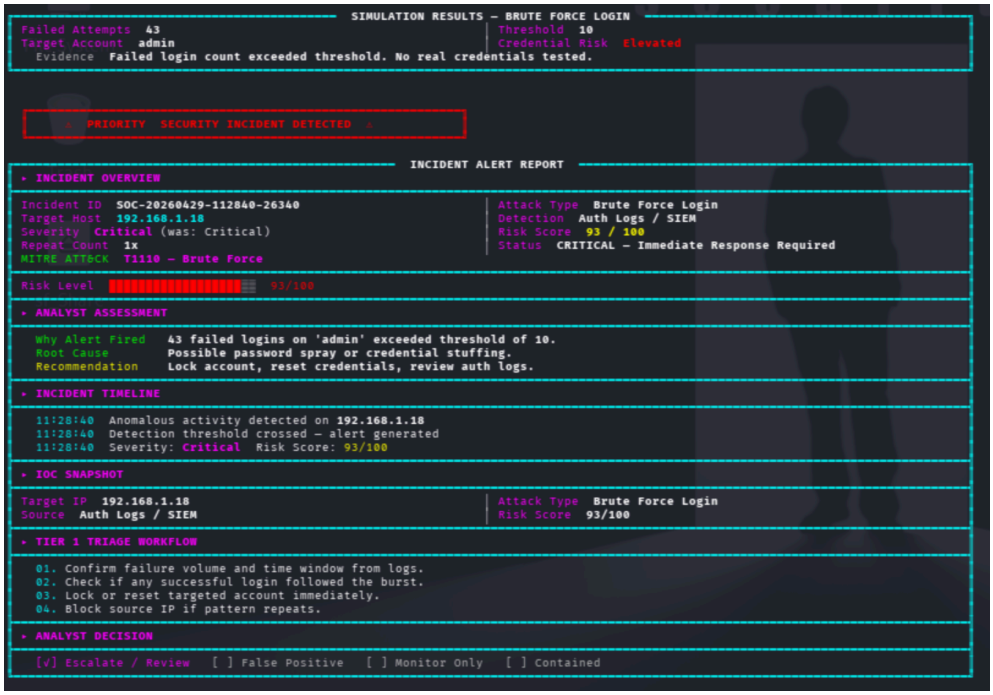


Figure 13. *Shows the Brute Force Login alert generated by CyberShield Simulator.*

```
}sim_brute_force() {
    local target="$1" fails sev
    fails=$((10 + RANDOM % 40))
    sev="High"; [ "$fails" -gt 30 ] && sev="Critical"
    status_warn "Brute Force Simulation on ${C_WHITE}${target}${NC}"
    ui_loading_bar "Analyzing failed authentication on ${target}" 22
    ui_box_top "$UI_WIDTH" "SIMULATION RESULTS — BRUTE FORCE LOGIN"
    ui_metric_row "$UI_WIDTH" "Failed Attempts"  "${C_WHITE}${fails}${NC}"  "Threshold"       "${C_WHITE}10${NC}"
    ui_metric_row "$UI_WIDTH" "Target Account"   "${C_WHITE}admin${NC}"     "Credential Risk" "${C_DANGER}Elevated${NC}"
    ui_box_line "$UI_WIDTH" "  ${C_GRAY}Evidence${NC}  ${C_WHITE}Failed login count exceeded threshold. No real credentials tested.
    ui_box_bottom "$UI_WIDTH"
    create_soc_alert "Brute Force Login" "$target" "$sev" \
        "Auth Logs / SIEM" "Escalated — Tier 2" "T1110 — Brute Force" \
        "${fails} failed logins on 'admin' exceeded threshold of 10." \
        "Possible password spray or credential stuffing." \
        "Lock account, reset credentials, review auth logs." \
        "Confirm failure volume and time window from logs." \
        "Check if any successful login followed the burst." \
        "Lock or reset targeted account immediately." \
        "Block source IP if pattern repeats."
-}
```

Code Snippet 6. *Shows the function used for simulating Brute Force detection and generating the SOC*

5.4. Malware Detection Simulation

Malware can cause serious damage to systems and data. This module simulates malicious activity and generates a high-severity SOC alert. It demonstrates how analysts identify and respond to critical threats quickly.

Figure 14. *Shows the Malware Detection alert generated by CyberShield Simulator.*

```
sim_malware() {
    local target="$1" fake_hash="f2a1b9c8d7e6a5f403928171abcdef12"
    status_warn "Malware Detection Simulation on ${C_WHITE}${target}${NC}"
    ui_loading_bar "Scanning file system for suspicious executables on ${target}" 22
    ui_box_top "$UI_WIDTH" "SIMULATION RESULTS — MALWARE DETECTION"
    ui_metric_row "$UI_WIDTH" "Suspicious File"  "${C_WHITE}/tmp/update_service.bin${NC}"    "Hash"      "${C_WHITE}${fake_hash}${N
    ui_metric_row "$UI_WIDTH" "Detection Type"   "${C_DANGER}Suspicious executable${NC}"    "Behavior"  "${C_WHITE}Possible staging
    ui_box_line "$UI_WIDTH" "  ${C_GRAY}Note${NC}  ${C_WHITE}No malware executed or dropped. Simulated detection only.${NC}"
    ui_box_bottom "$UI_WIDTH"
    create_soc_alert "Malware Detection" "$target" "Critical" \
        "Endpoint Detection / EDR" "CRITICAL — Immediate Response" "T1204 — User Execution" \
        "Suspicious executable at /tmp/update_service.bin with anomalous hash." \
        "Possible malware staging, dropper, or persistence mechanism." \
        "Isolate host, collect evidence, escalate to incident response." \
        "Immediately isolate the affected host from the network." \
        "Collect file path, hash, and running process evidence." \
        "Run full endpoint scan and check persistence paths." \
        "Escalate to incident response team for containment."
}
```

Code Snippet 8. *The function used for simulating Malware detection and generating the SOC alert.*

6. Dashboard and Log Management

6.1 SOC Executive Dashboard

The dashboard shows important information such as detected hosts, attack type, severity level, MITRE ATT&CK mapping, and investigation status. This makes incident response faster and improves visibility during security monitoring.

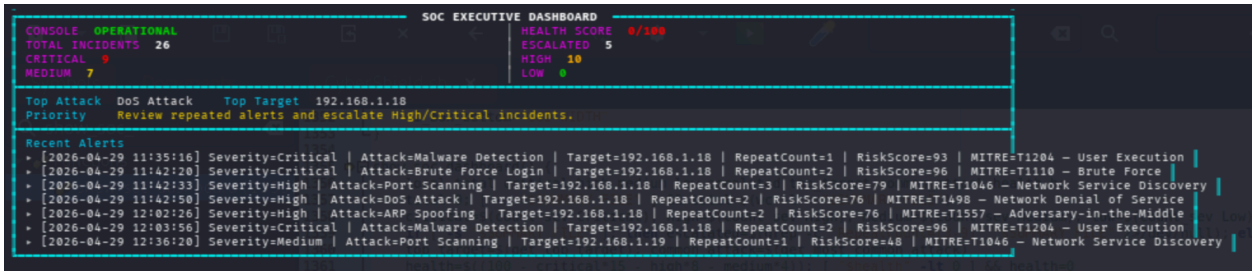


Figure 15. *SOC Executive Dashboard showing alerts, risk score, and incident status.*

```
show_soc_dashboard() {
    local total critical high medium low escalated top_target common_attack health hc
    total=0; [ -s "$SUMMARY_FILE" ] && total=$(wc -l < "$SUMMARY_FILE")
    critical=$(count_sev Critical); high=$(count_sev High); medium=$(count_sev Medium); low=$(count_sev Low)
    if [ -s "$SUMMARY_FILE" ]; then escalated=$(grep -c 'RepeatCount=[3-9]' "$SUMMARY_FILE" 2>/dev/null); else escalated=0; fi
    top_target=$(get_top_target); common_attack=$(get_most_common_attack)
    health=$((100 - critical*15 - high*8 - medium*4)); [ "$health" -lt 0 ] && health=0
    hc="$C_GREEN"; [ "$health" -lt 70 ] && hc="$C_YELLOW"; [ "$health" -lt 40 ] && hc="$C_RED"
    ui_topbar
    ui_box_top "$UI_WIDTH" "SOC EXECUTIVE DASHBOARD"
    ui_metric_row "$UI_WIDTH" "CONSOLE" "${C_GREEN}OPERATIONAL${NC}" "HEALTH SCORE" "${hc}${health}/100${NC}"
    ui_metric_row "$UI_WIDTH" "TOTAL INCIDENTS" "$total" "ESCALATED" "$escalated"
    ui_metric_row "$UI_WIDTH" "CRITICAL" "${C_RED}${critical}${NC}" "HIGH" "${C_ORANGE}${high}${NC}"
    ui_metric_row "$UI_WIDTH" "MEDIUM" "${C_YELLOW}${medium}${NC}" "LOW" "${C_GREEN}${low}${NC}"
    ui_box_separator "$UI_WIDTH"
    ui_box_line "$UI_WIDTH" "${C_NEON}Top Attack${NC}  ${common_attack}    ${C_NEON}Top Target${NC}  ${top_target}"
    ui_box_line "$UI_WIDTH" "${C_NEON}Priority${NC}    ${C_YELLOW}Review repeated alerts and escalate High/Critical incidents.${NC}"
    ui_box_separator "$UI_WIDTH"
    ui_box_line "$UI_WIDTH" "${C_NEON}Recent Alerts${NC}"
    if [ -s "$SUMMARY_FILE" ]; then
        tail -n 7 "$SUMMARY_FILE" | while IFS= read -r line; do ui_box_line "$UI_WIDTH" "${C_GRAY}▸${NC} $line"; done
    else
        ui_box_line "$UI_WIDTH" "${C_GRAY}No incidents yet. Run a simulation first.${NC}"
    fi
    ui_box_bottom "$UI_WIDTH"
}
```

Code Snippet 9: *Function used to display the SOC dashboard and incident summary.*

6.2 Alert Generation and Severity Classification

Severity classification helps SOC analysts prioritize incidents effectively.

- Low-level alerts require monitoring
- Medium-level alerts require investigation
- High and Critical alerts require immediate response

This classification improves decision-making and ensures efficient incident handling.

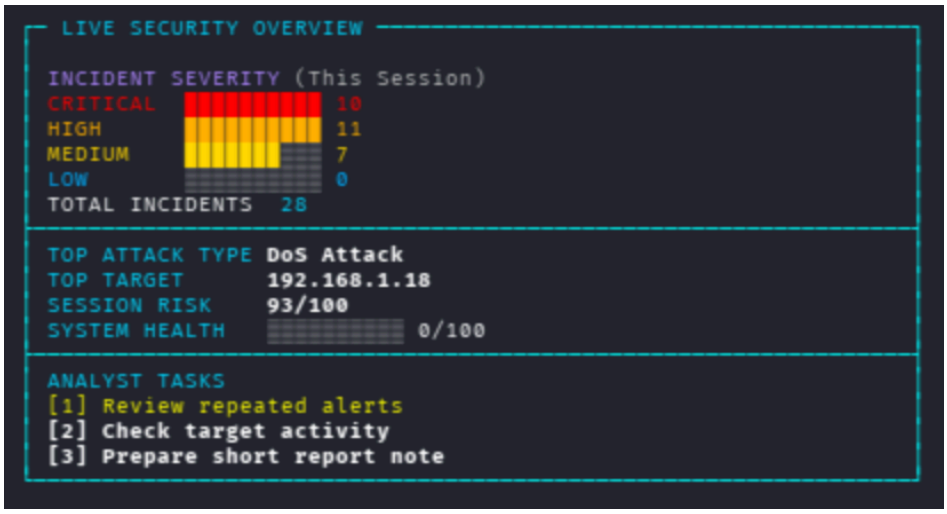


Figure 15.*Alert severity classification showing Low, Medium, High, and Critical levels.*

```
create_soc_alert() {
    local attack="$1" target="$2" sev="$3" src="$4" status="$5" mitre="$6"
    local why="$7" cause="$8" action="$9" t1="${10}" t2="${11}" t3="${12}" t4="${13}"
    local id rc rcount orig_sev risk
    id=$(generate_incident_id)
    rc=$(get_repeat_count "$attack" "$target")
    rcount=$((rc+1))
    orig_sev="$sev"
    sev=$(escalate_severity "$sev" "$rcount")
    status=$(get_alert_status "$sev" "$rcount")
    risk=$(calculate_risk_score "$sev" "$rcount")
    if [ "$rcount" -ge 3 ]; then
        why="$why [Repeated: ${rcount}x]"
        cause="$cause Correlation engine flagged repeated activity."
        action="$action PRIORITY: escalate immediately."
    fi
    print_soc_alert "$id" "$attack" "$target" "$sev" "$src" "$status" "$mitre" \
        "$why" "$cause" "$action" "$t1" "$t2" "$t3" "$t4" "$rcount" "$risk" "$orig_sev"
    save_soc_record "$id" "$attack" "$target" "$sev" "$src" "$status" "$mitre" \
        "$why" "$cause" "$action" "$t1" "$t2" "$t3" "$t4" "$rcount" "$risk" "$orig_sev"
    record_correlation_event "$attack" "$target" "$sev" "$id"
    SESSION_LAST_SEVERITY="$sev"
    SESSION_LAST_RISK="$risk"
    echo ""
    status_ok "Incident saved → ${C_WHITE}${LOG_FILE}${NC}"
    status_ok "IOC exported  → ${C_WHITE}${IOC_FILE}${NC}"
}
```

Code Snippet 10: *Function used to generate alerts and assign severity levels.*

6.3 Incident Logging and IOC Export Testing and Results

All simulation results are automatically saved inside **/var/log/cybershield/** to support reporting and forensic investigation. The system stores incidents, reports, summaries, correlation logs, and IOC export files for analyst review.

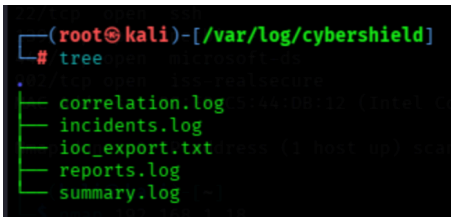


Figure 17. *CyberShield log files stored in* ***/var/log/cybershield/.***

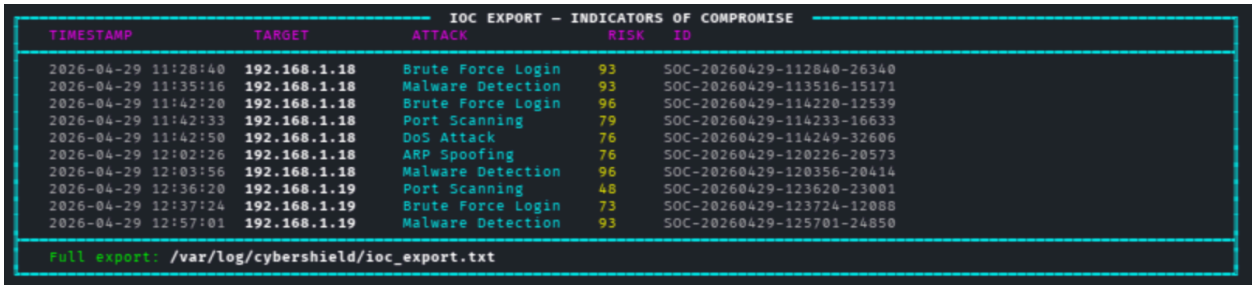


Figure 18: *IOC export showing attack records and risk scores.*

```
show_ioc_export() {
    echo ""
    ui_box_top "$UI_WIDTH" "IOC EXPORT — INDICATORS OF COMPROMISE"
    if [ ! -s "$IOC_FILE" ]; then
        ui_box_line "$UI_WIDTH" "  ${C_GRAY}No IOCs recorded yet.${NC}"
        ui_box_bottom "$UI_WIDTH"
        return
    fi
    ui_box_line "$UI_WIDTH" "  ${C_ACCENT}TIMESTAMP            TARGET          ATTACK              RISK   ID${NC}"
    ui_box_separator "$UI_WIDTH"
    tail -n 10 "$IOC_FILE" | while IFS= read -r line; do
        local ts tgt atk risk iid
        ts=$(echo "$line"  | sed -n 's/^\[\([^]]*\)\].*/\1/p')
        tgt=$(echo "$line" | sed -n 's/.*target=\([^|]*\)|.*/\1/p'  | sed 's/ *$//')
        atk=$(echo "$line" | sed -n 's/.*attack=\([^|]*\)|.*/\1/p'  | sed 's/ *$//')
        risk=$(echo "$line" | sed -n 's/.*risk=\([^|]*\)|.*/\1/p'   | sed 's/ *$//')
        iid=$(echo "$line"  | sed -n 's/.*id=\(.*\)/\1/p'           | sed 's/ *$//')
        ui_box_line "$UI_WIDTH" "  ${C_GRAY}$(printf '%-21s' "$ts")${NC}${C_WHITE}$(printf '%-17s' "$tgt")${NC}${C_PRIMARY}$(printf '%-21s' "$atk")${NC}${C_WARN}$(pr
    done
    ui_box_separator "$UI_WIDTH"
    ui_box_line "$UI_WIDTH" "  ${C_SUCCESS}Full export:${NC} ${C_WHITE}${IOC_FILE}${NC}"
    ui_box_bottom "$UI_WIDTH"
}
```

Code Snippet 11:*Function used to save logs and export IOC files.*

7. Testing and Results

The system was tested in a controlled lab environment using Kali Linux and an Ubuntu host. Each attack module was verified to ensure proper alert generation, risk scoring, logging, and dashboard updates.

- Port scanning was validated through TCP SYN packets
- DoS simulation was confirmed using repeated ICMP traffic
- Brute force and malware simulations were verified through generated alerts

The results confirm that the system functions correctly while remaining safe for educational use.

7.1 Attack Simulation Testing

I tested Port Scanning, DoS Attack, Brute Force Login Detection, and Malware Detection using Kali Linux VM and Ubuntu host in a controlled environment.

Port scanning validation was confirmed using Wireshark by observing repeated TCP SYN packets. DoS simulation was validated using repeated ICMP traffic between Kali and Ubuntu. Brute Force and Malware Detection were verified through generated SOC alerts and dashboard responses.

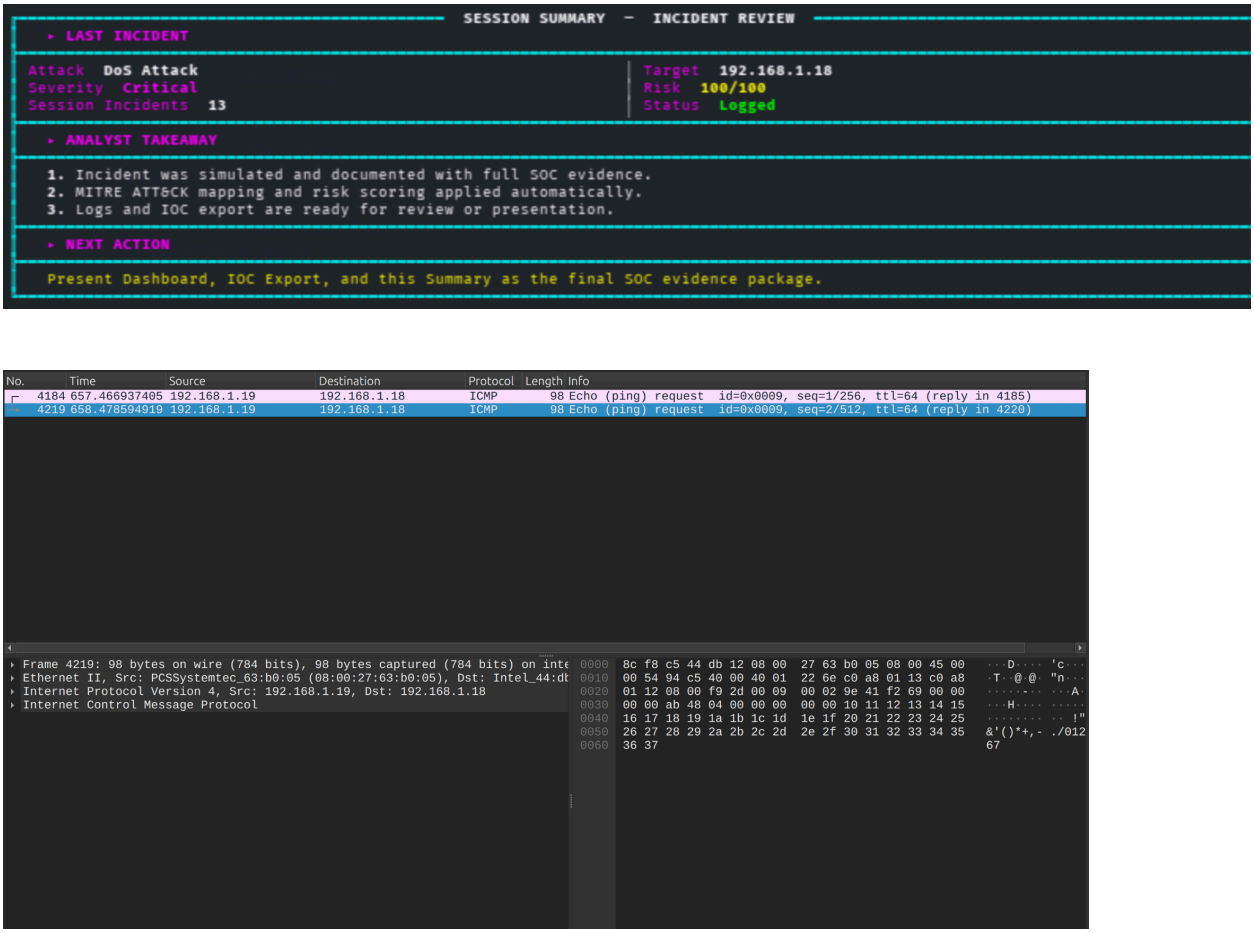


Figure 19. *Wireshark validation showing repeated ICMP traffic during DoS Attack simulation between Kali Linux and the Ubuntu target system.*

7.2 Dashboard and Log Verification

The dashboard was reviewed to confirm accurate display of:

- Incident IDs
- Severity levels
- Risk scores
- MITRE ATT&CK mapping

Additionally, the **/var/log/cybershield/** directory was verified to ensure all logs, reports, and IOC exports were correctly generated after each simulation.

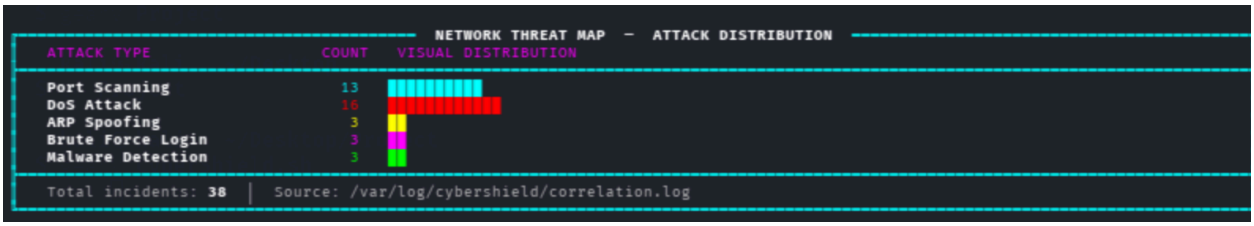


Figure 19. *Network threat map for attack simulator.*

8. Security Considerations and Future Improvements

CyberShield Simulator is designed for safe educational use and does not execute real attacks or harmful operations. Future improvements may include:

- Integration with SIEM tools such as Splunk
- Real-time alert monitoring
- Threat intelligence integration
- Web-based dashboard interface

9. Conclusion

CyberShield Simulator successfully demonstrates how a Security Operations Center operates in detecting, analyzing, and responding to cyber threats. The project

integrates key SOC components, including network scanning, attack simulation, alert generation, severity classification, dashboard monitoring, and incident logging into a single platform.

Through this project, practical understanding of SOC workflows, incident response, and cybersecurity monitoring has been achieved. The system provides a strong foundation for future development and can be extended to integrate with real-world tools such as SIEM platforms.

10. References

- MITRE ATT&CK. ATT&CK Framework for Enterprise Security Detection and Response. Available at: https://attack.mitre.org/
- Nmap. Official Documentation for Network Discovery and Security Auditing. Available at: https://nmap.org/
- Wireshark. Official Documentation for Packet Capture and Traffic Analysis. Available at: https://www.wireshark.org/
- Kali Linux. Official Documentation and Penetration Testing Platform. Available at: https://www.kali.org/
- GNU Bash Documentation. Bash Reference Manual for Shell Scripting and Automation. Available at: https://www.gnu.org/software/bash/manual/
- Incident Response Lifecycle. Security Operations Center (SOC) Analyst Workflow and Triage Process.
- Linux Log Management Documentation. Using **/var/log** for Security Monitoring and Incident Investigation.
- Endpoint Detection and Response (EDR) Concepts for Malware Detection and IOC Analysis.